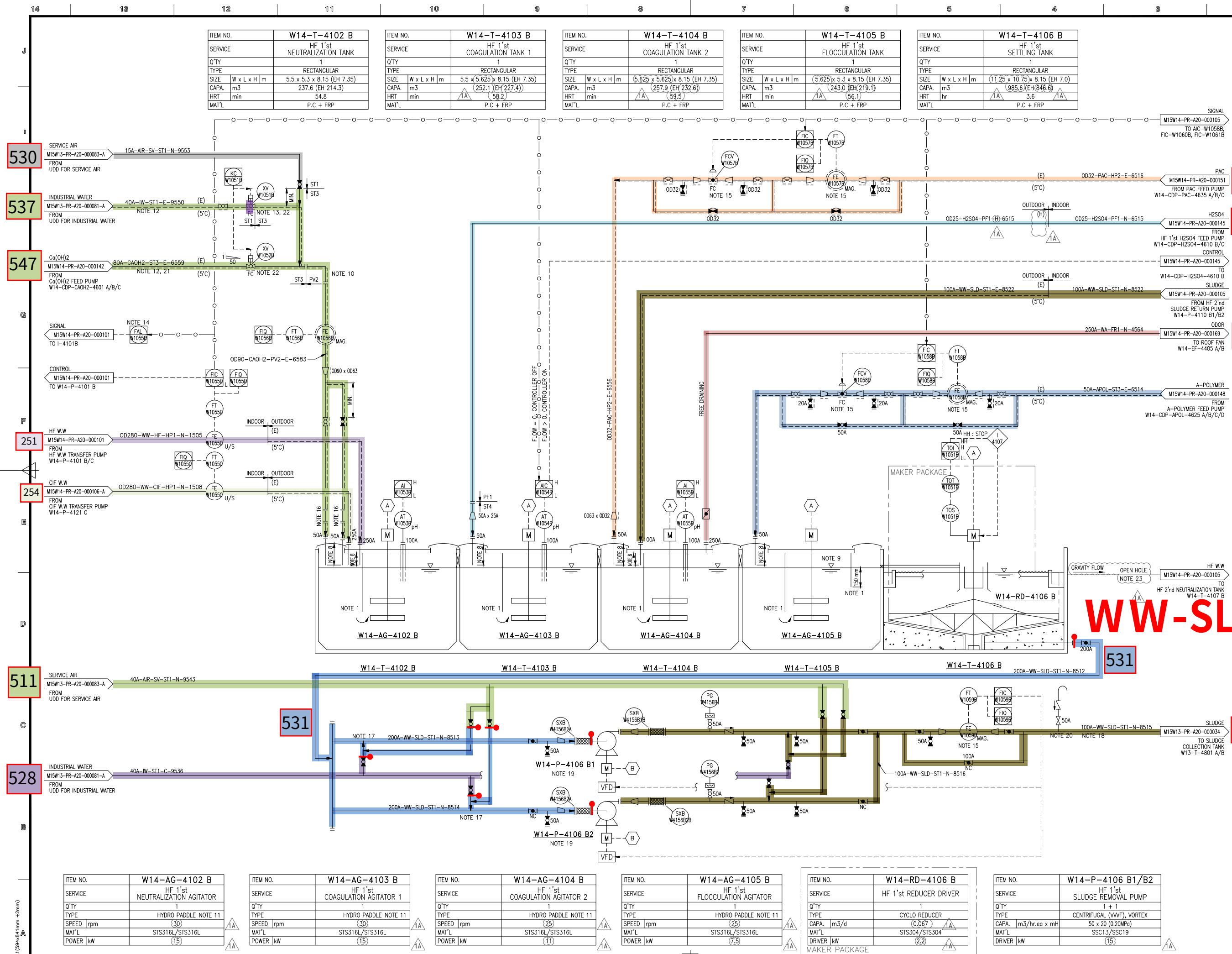
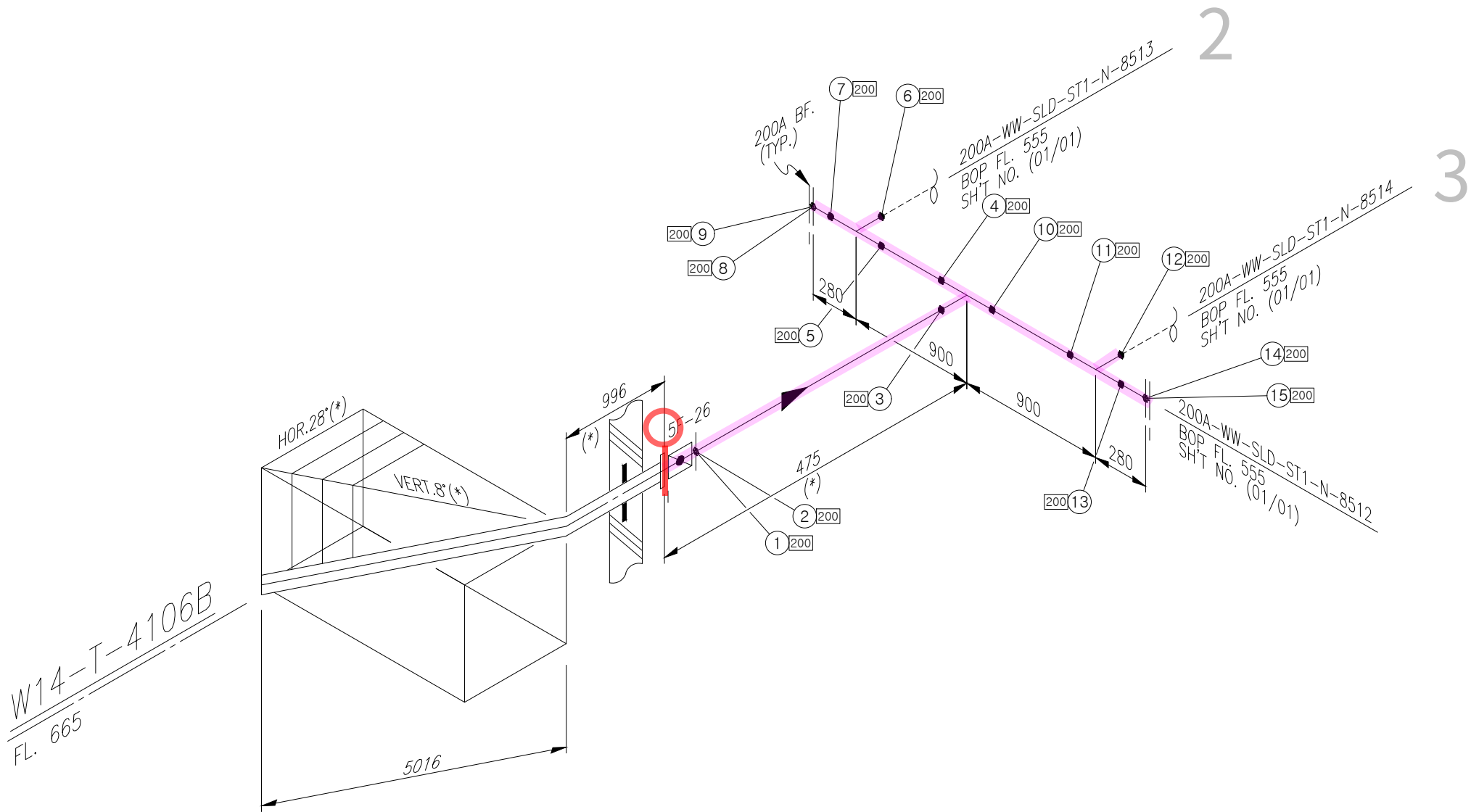
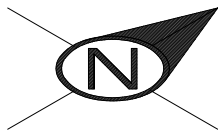


- NOTE**
1. Baffle 제형 : FRP  
Baffle 상부에 0.02 m2 이상 AIR HOLE 설치.
  2. NK와 HWL 이하로 연결되는 모든 배관은 BLOCK 밸브 1개 이상.
  3. 3.2m 이상 높은 곳에 설치된 밸브는 CHAIN VALVE 또는 PLATFORM 구성할 것.
  4. INHOLE : 2개 이상 설치한다.  
INHOLE COVER : FRP  
시 CLOSE 되는 배관에는 LOTO 설치.
  5. CENTRIFUGAL PUMP.  
MAT'L 표기 : CASING / IMPELLER
  6. AGITATOR MAT'L 표기 : SHAFT/BLADE
  7. TANK 내 유입시, 육안 확인이 가능하고, 비산이 되지 않도록 최소 길이로 변경 할 것.  
B 으로부터 100mm 떨어질 것
  8. FRP LINING TANK의 SLEEVE는 504 + FRP LINING 적용.
  9. 10. Co(OH)2 배관의 ELBOW 부분의 FLANGE 설치할 것
  11. 주용 VENDOR에서 결정한다.
  12. 라인은 상부에서 시공 시 유입하드록 설치한다.
  13. W1051B OPEN (12시간마다 15초) 일 경우 : W1052B CLOSE
  14. W14-P-4101 B가 RUN 후 30초 후부터 동작한다.
  15. INSTRUMENT VENDOR 선정 후 CONTROL SCHEME 및 CONTROL VALVE INLET/OUTLET NOZZLE SIZE 확정
  16. 구간 투명 PVC 적용. (주입 여부 확인용)
  17. SHING 용 WATER / AIR 배관은 45도 서선으로 연결할 것.
  18. SLUDGE REMOVAL PUMP를 배관은 PUMP SUCTION 속의 TANK 수위보다 HIGH POINT 형성.
  19. SLUDGE PUMP는 TIMER에 의해 운전됨.
  20. HIGH POINT에 SIPHON BREAKER 설치할것.  
(CHECK VALVE의 경우 DUAL PLATE TYPE으로 적용)
  21. 소액의 공금 HEADER에서 주입되는 POINT 별 ISOLATION VALVE, ON-OFF VALVE, FLUSHING LINE, SPEC. BREAK 까지 MIN. DISTANCE 적용하여, 역침 최소화 및 주입배관 고체를 용이하게 한다.
  22. XPM SPEED CONTROLLER (ACCESSORY) 설치.
  23. CONCRETE WALL OPENING을 통한 유체 이용 OPENING SIZE IS 800mm x 400mm

|             |                           |          |
|-------------|---------------------------|----------|
| 1A          | REVISED AFC               | 25.08.14 |
| 1           | REVISED AFC               | 25.01.17 |
| 0           | APPROVAL FOR CONSTRUCTION | 24.09.30 |
| 수정          | 실명                        | 날짜       |
| REV.        | DESCRIPTION               | DATE     |
| 제도          | 이인학                       |          |
| DRAWN BY    |                           |          |
| 토           | 한진희                       |          |
| DESIGNED BY |                           |          |
| 인           | 박동명                       |          |
| CHECKED BY  |                           |          |
| 제작사         |                           |          |

|                                            |  |                         |
|--------------------------------------------|--|-------------------------|
| SK 에코플랜트                                   |  |                         |
| 공사명                                        |  |                         |
| PROJECT TITLE                              |  |                         |
| M15X Ph4 WWT Project                       |  |                         |
| 도면명                                        |  |                         |
| DWG. TITLE                                 |  |                         |
| PIPING & INSTRUMENT DIAGRAM                |  |                         |
| HF 1st COAGULATION & SETTLING TANK B (2/2) |  |                         |
| 속척                                         |  |                         |
| SCALE                                      |  |                         |
| NONE                                       |  |                         |
| 날짜                                         |  |                         |
| DATE                                       |  |                         |
| 2025. 08. 14                               |  |                         |
| PROJECT NO.                                |  | 22330D                  |
| 도면번호                                       |  |                         |
| SHEET NO.                                  |  | CJ-M15W14-PR-A20-000104 |
|                                            |  | 1A                      |



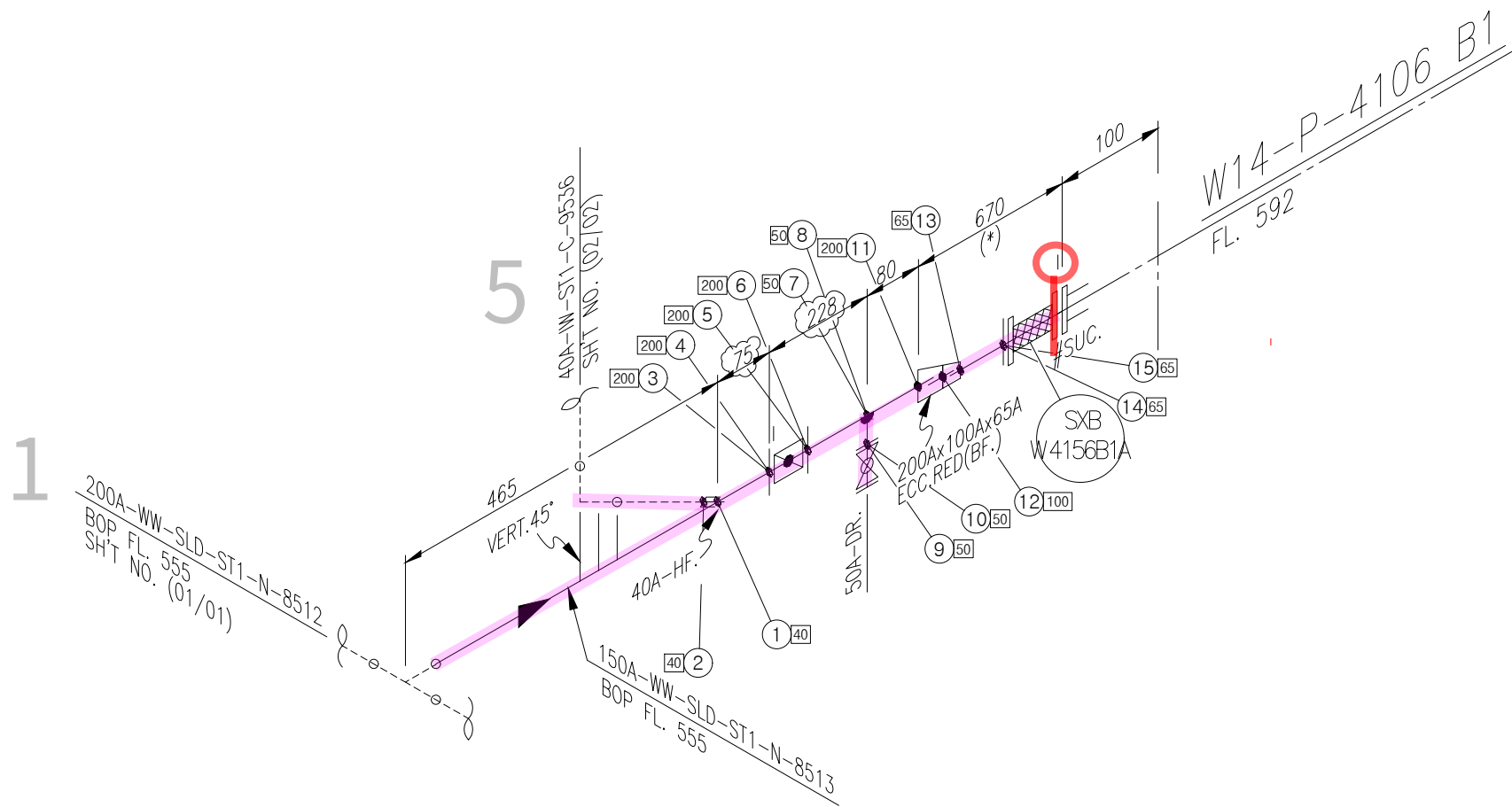
<NOTE>  
1. (\*) FIELD ADJUST

\* PIPE : P/P1 \* 90° ELBOW : E90 \* 45° ELBOW : E45 \* TEE : T \* RED.TEE : TR \* STUB-IN : STU \* REINFORCING PAD : RP \* CON.RED : RC \* ECC.RED : RE \* CON.SWAGE : RCS/RCS1 \* ECC.SWAGE : RES/RES1 \* NIPPLE : NP/NP1/NP2 \* CAP : CAP/CAP1 \* PLUG : PLG \* FULL COUPLING : FC \* HALF COUPLING : HF/HF1  
\* STUB-IN : STU \* SOCKET(HDPE/PVC) : SC \* SADDLE(HDPE) : SD \* FLANGE : F/F1 \* GASKET : G/G1/G2 \* BOLT & NUT : B/B1 \* BLIND FLANGE : FB/FB1 \* GATE VA. : VG/VG1/VG2 \* BALL VA. : VB/VB1/VB2/VB2LO \* BUTTERFLY VA. : VF/VF4 \* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7  
\* GLOBE VA. : VL/VL1/VL2 \* DIAPHRAGM VA. : VD/VDLO \* DAMPER : VP \* QUICK COUPLING : SQC \* FLEXIBLE JOINT(STEEL) : FLXB \* FLEXIBLE JOINT(RUBBER) : FLXR \* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL \* FLEXIBLE HOSE : FLXH \* STEAM TRAP : ST \* SPRAY NOZZLE : SPN \* EMERGENCY SHOWER : SSH  
\* STRAINER(Y-TYPE) : YSTR \* STRAINER(T-TYPE) : TSTR \* STRAINER(BUCKET-TYPE) : BSTR \* ANCHOR PLATE : STOP \* BIRD SCREEN : BS \* ISOLATION KIT : IK \* UNLOADING BOX : UB \* AIR TRAP : AT \* LOTO : LOTO \* SIGHT GLASS : SG \* PRESSURE GAUGE : PG \* SHILED : SH \* SPECIAL BOLT & NUT : BL/BL1

|            |        |   |           |                     |               |                                             |     |              |          |                           |     |     |       |                                                                                                                        |                        |                     |           |
|------------|--------|---|-----------|---------------------|---------------|---------------------------------------------|-----|--------------|----------|---------------------------|-----|-----|-------|------------------------------------------------------------------------------------------------------------------------|------------------------|---------------------|-----------|
| INSULATION | CODE   | N | CONDITION | PRESS.<br>(kg/cm2g) | TEMP.<br>(°C) | AREA                                        | W14 |              |          |                           |     |     |       | <div>CONTRACTOR</div> <div></div> | PROJECT NAME           |                     |           |
|            | THICK. | — | DESIGN    | 0.8                 | 60            | UNIT                                        | 5FA |              |          |                           |     |     |       |                                                                                                                        | M15X Ph4 Project       |                     |           |
|            |        |   | OPERATING | ATM                 | AMB           | REF. PLAN DWG.<br>CJ-M15W14-PI-A30-050006-A |     |              |          |                           |     |     |       | COMMODITY                                                                                                              | WW-SLD                 | JOB NO.             |           |
|            |        |   |           |                     |               | REF. P & ID<br>CJ-M15W14-PR-A20-000104      |     | <div>△</div> | 25.01.24 | APPROVED FOR CONSTRUCTION | 황래중 | 안남후 | 박동영   | LINE NO.                                                                                                               | 200A-WW-SLD-ST1-N-8512 | SH'T NO.<br>01 / 01 | REV.<br>0 |
|            |        |   |           |                     |               |                                             |     | REV. NO.     | DATE     | DESCRIPTION               | BY  | CHK | APP'D |                                                                                                                        |                        |                     |           |



| ITEM | SIZE      | Q'TY | DESCRIPTION                                       |
|------|-----------|------|---------------------------------------------------|
| ST1  |           |      |                                                   |
| P    | 200A      | 0.5M | PIPE, STS 304TP, ERW, BE – SCH 10S                |
| P    | 65A       | 0.3M | PIPE, STS 304TP, ERW, BE – SCH 10S                |
| P    | 50A       | 0.1M | PIPE, STS 304TP, ERW, BE – SCH 10S                |
| RE   | 200Ax100A | 1EA  | REDUCER ECC, STS304, ERW, BE – SCH 10S            |
| RE   | 100Ax65A  | 1EA  | REDUCER ECC, STS304, ERW, BE – SCH 10S            |
| STU  | 200Ax50A  | 1EA  | STUB-IN BRANCH                                    |
| HF   | 40A       | 1EA  | COUPLING HALF, STSF304, SW – SCH 10S              |
| VB   | 50A       | 1EA  | BALL VALVE, KS 10K, RF FLG'D, BODY-SSC13 /        |
|      |           |      | TRIM-STS304 / SEAT-PTFE, FB, LEVER                |
| VF   | 200A      | 1EA  | BUTTERFLY VALVE, KS 10K, WAFER RF, BODY-GCD450 /  |
|      |           |      | TRIM-STS304 / EPDM-LINED, LEVER                   |
| F    | 200A      | 2EA  | FLG SO, KS 10K, RF, STS304                        |
| F    | 65A       | 1EA  | FLG SO, KS 10K, RF, STS304                        |
| F    | 50A       | 1EA  | FLG SO, KS 10K, RF, STS304                        |
| FB   | 50A       | 1EA  | FLG BLIND, KS 10K, RF, STS304                     |
| G    | 200A      | 2EA  | GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K         |
| G    | 65A       | 2EA  | GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K         |
| G    | 50A       | 2EA  | GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K         |
| B    | 65A       | 2SET | MACHINE BOLT & NUTS :STS 304 / W WASHERS          |
| B    | 50A       | 2SET | MACHINE BOLT & NUTS :STS 304 / W WASHERS          |
| BL   | 200A      | 1SET | MACHINE BOLT & NUTS :STS 304 / W WASHERS(SPECIAL) |
| FLXB | 65A       | 1EA  | FLEXIBLE JOINT(TPC STEEL TYPE)                    |



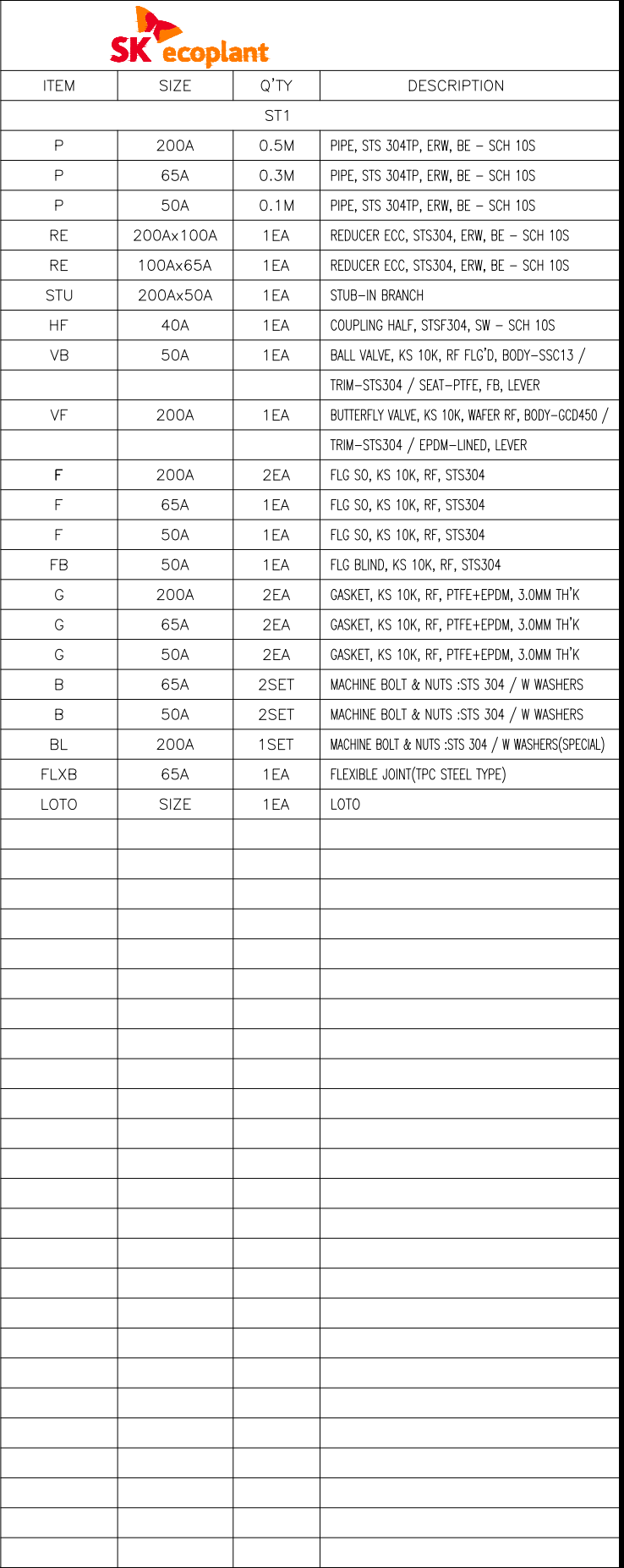
<NOTE>

### 1. (\*) FIELD ADJUST

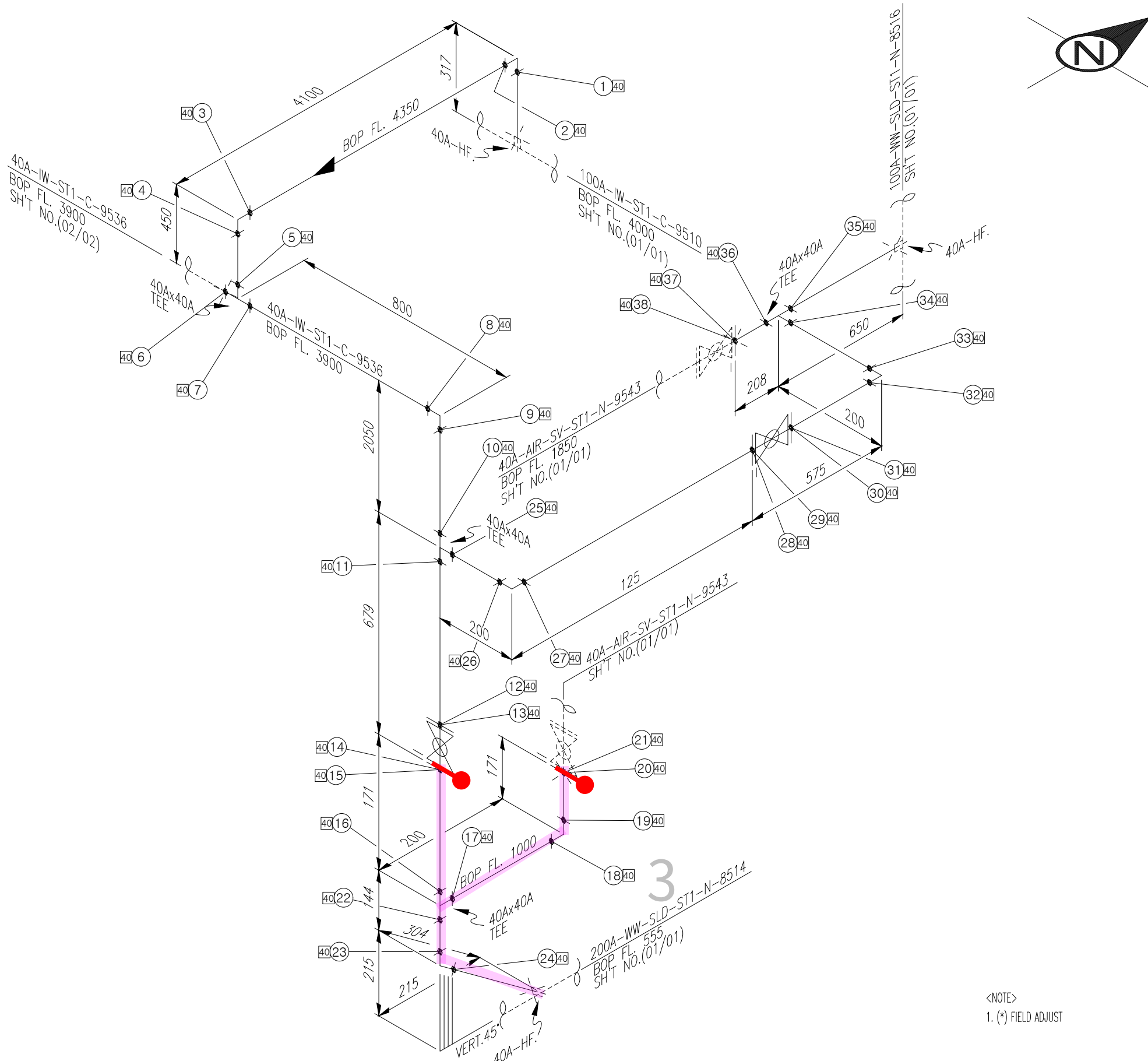
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|---------------------------|---------------------------|--------------------------------|------------------------|--------------------------------|---------------------------------|-------------------------------------------|-------------------------|------------------------------------|--------------------------|----------------------------------------------|-----------------------|-------------------------------|--------------|----------------------|--------------------------|
| * PIPE : P/P1             | * 90° ELBOW : E90         | * 45° ELBOW : E45              | * TEE : T              | * RED.TEE : TR                 | * STUB-IN : STU                 | * REINFORCING PAD : RP                    | * CON.RED : RC          | * ECC.RED : RE                     | * CON.SWAGE : RCS/RCS1   | * ECC.SWAGE : RES/RES1                       | * NIPPLE : NP/NP1/NP2 | * CAP : CAP/CAP1              | * PLUG : PLG | * FULL COUPLING : FC | * HALF COUPLING : HF/HF1 |
| * STUB-IN : STU           | * SOCKET(HDPE/PVC) : SC   | * SADDLE(HDPE) : SD            | * FLANGE : F/F1        | * GASKET : G/G1/G2             | * BOLT & NUT : B/B1             | * BLIND FLANGE : FB/FB1                   | * GATE VA. : VG/VG1/VG2 | * BALL VA. : VB/VB1/VB10/VB2/VB210 | * BUTTERFLY VA. : VF/VF4 | * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7 |                       |                               |              |                      |                          |
| * GLOBE VA. : VL/VL1/VL2  | * DIAPHRAGM VA. : VD/VD10 | * DAMPER : VP                  | * QUICK COUPLING : SQC | * FLEXIBLE JOINT(STEEL) : FLXB | * FLEXIBLE JOINT(RUBBER) : FLXR | * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL | * FLEXIBLE HOSE : FLXH  | * STEAM TRAP : ST                  | * SPRAY NOZZLE : SPN     | * EMERGENCY SHOWER : SSH                     |                       |                               |              |                      |                          |
| * STRAINER(Y-TYPE) : YSTR | * STRAINER(T-TYPE) : TSTR | * STRAINER(BUCKET-TYPE) : BSTR | * ANCHOR PLATE : STOP  | * BIRD SCREEN : BS             | * ISOLATION KIT : IK            | * UNLOADING BOX : UB                      | * AIR TRAP : AT         | * LOTO : LOTO                      | * SIGHT GLASS : SG       | * PRESSURE GAUGE : PG                        | * SHILED : SH         | * SPECIAL BOLT & NUT : BL/BL1 |              |                      |                          |

|            |        |   |           |                     |               |                           |     |                                                                                       |          |                           |     |     |                                                                                       |            |                        |                     |           |
|------------|--------|---|-----------|---------------------|---------------|---------------------------|-----|---------------------------------------------------------------------------------------|----------|---------------------------|-----|-----|---------------------------------------------------------------------------------------|------------|------------------------|---------------------|-----------|
| INSULATION | CODE   | N | CONDITION | PRESS.<br>(kg/cm2g) | TEMP.<br>(°C) | AREA                      | W14 |                                                                                       |          |                           |     |     |  | CONTRACTOR | PROJECT NAME           |                     |           |
|            | THICK. | — | DESIGN    | 0.8                 | 60            | UNIT                      | 5FA |                                                                                       |          |                           |     |     |                                                                                       |            | M15X Ph4 Project       |                     |           |
|            |        |   | OPERATING | ATM                 | AMB           | REF. PLAN DWG.            |     |                                                                                       |          |                           |     |     | COMMODITY                                                                             | WW—SLD     | JOB NO.                |                     |           |
|            |        |   |           |                     |               | CJ—M15W14—PI—A30—050006—A |     |  | 25.01.24 | APPROVED FOR CONSTRUCTION | 황래중 | 안남후 |                                                                                       |            | 박동영                    | 22330D              |           |
|            |        |   |           |                     |               | REF. P & ID               |     | REV. NO.                                                                              | DATE     | DESCRIPTION               | BY  | CHK | APP'D                                                                                 | LINE NO.   | 200A—WW—SLD—ST1—N—8513 | SH'T NO.<br>01 / 01 | REV.<br>0 |
|            |        |   |           |                     |               | CJ—M15W14—PR—A20—000104   |     |                                                                                       |          |                           |     |     |                                                                                       |            |                        |                     |           |

경고 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음



|            |        |   |           |                     |               |                                             |                                                                                       |          |                           |     |     |       |                                                                                       |                        |                     |           |
|------------|--------|---|-----------|---------------------|---------------|---------------------------------------------|---------------------------------------------------------------------------------------|----------|---------------------------|-----|-----|-------|---------------------------------------------------------------------------------------|------------------------|---------------------|-----------|
| INSULATION | CODE   | N | CONDITION | PRESS.<br>(kg/cm2g) | TEMP.<br>(°C) | AREA W14                                    |                                                                                       |          |                           |     |     |       | CONTRACTOR                                                                            | PROJECT NAME           |                     |           |
|            | THICK. | — | DESIGN    | 0.8                 | 60            | UNIT 5FA                                    |                                                                                       |          |                           |     |     |       |  | M15X Ph4 Project       |                     |           |
|            |        |   | OPERATING | ATM                 | AMB           | REF. PLAN DWG.<br>CJ-M15W14-PI-A30-050006-A |                                                                                       |          |                           |     |     |       | COMMODITY                                                                             | WW—SLD                 | JOB NO.             |           |
|            |        |   |           |                     |               | REF. P & ID<br>CJ-M15W14-PR-A20-000104      |  | 25.01.24 | APPROVED FOR CONSTRUCTION | 황래중 | 안남후 | 박동영   | LINE NO.                                                                              | 200A-WW-SLD-ST1-N-8514 | SH'T NO.<br>01 / 01 | REV.<br>0 |
|            |        |   |           |                     |               |                                             | REV. NO.                                                                              | DATE     | DESCRIPTION               | BY  | CHK | APP'D |                                                                                       |                        |                     |           |



<NOTE>  
1. (\*) FIELD ADJUST

\* PIPE : P/P1   \* 90° ELBOW : E90   \* 45° ELBOW : E45   \* TEE : T   \* RED.TEE : TR   \* STUB-IN : STU   \* REINFORCING PAD : RP   \* CON.RED : RC   \* ECC.RED : RE   \* CON.SWAGE : RCS/RCS1   \* ECC.SWAGE : RES/RES1   \* NIPPLE : NP/NP1/NP2   \* CAP : CAP/CAP1   \* PLUG : PLG   \* FULL COUPLING : FC   \* HALF COUPLING : HF/HF1  
\* STUB-IN : STU   \* SOCKET(HDPE/PVC) : SC   \* SADDLE(HDPE) : SD   \* FLANGE : F/F1   \* GASKET : G/G1/G2   \* BOLT & NUT : B/B1   \* BLIND FLANGE : FB/FB1   \* GATE VA. : VG/VG1/VG2   \* BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO   \* BUTTERFLY VA. : VF/VF4   \* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7  
\* GLOBE VA. : VL/VL1/VL2   \* DIAPHRAGM VA. : VD/VDLO   \* DAMPER : VP   \* QUICK COUPLING : SQC   \* FLEXIBLE JOINT(STEEL) : FLXB   \* FLEXIBLE JOINT(RUBBER) : FLXR   \* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL   \* FLEXIBLE HOSE : FLXH   \* STEAM TRAP : ST   \* SPRAY NOZZLE : SPN   \* EMERGENCY SHOWER : SSH  
\* STRAINER(Y-TYPE) : YSTR   \* STRAINER(T-TYPE) : TSTR   \* STRAINER(BUCKET-TYPE) : BSTR   \* ANCHOR PLATE : STOP   \* BIRD SCREEN : BS   \* ISOLATION KIT : IK   \* UNLOADING BOX : UB   \* AIR TRAP : AT   \* LOTO : LOTO   \* SIGHT GLASS : SG   \* PRESSURE GAUGE : PG   \* SHILED : SH   \* SPECIAL BOLT & NUT : BL/BL1

|            |        |    |           |                     |               |                                             |     |                                                                                     |          |                           |     |     |       |            |                                                                                       |                     |           |
|------------|--------|----|-----------|---------------------|---------------|---------------------------------------------|-----|-------------------------------------------------------------------------------------|----------|---------------------------|-----|-----|-------|------------|---------------------------------------------------------------------------------------|---------------------|-----------|
| INSULATION | CODE   | C  | CONDITION | PRESS.<br>(kg/cm2g) | TEMP.<br>(°C) | AREA                                        | W14 |                                                                                     |          |                           |     |     |       | CONTRACTOR |  | PROJECT NAME        |           |
|            | THICK. | 25 | DESIGN    | 13.5                | 60            | UNIT                                        | 5FA |                                                                                     |          |                           |     |     |       |            |                                                                                       | M15X Ph4 Project    |           |
|            |        |    | OPERATING | 10                  | AMB           | REF. PLAN DWG.<br>CJ-M15W14-PI-A30-050006-C |     |                                                                                     |          |                           |     |     |       | COMMODITY  | IW                                                                                    | JOB NO.             |           |
|            |        |    |           |                     |               | REF. P & ID<br>CJ-M15W14-PR-A20-000104      |     |  | 25.01.24 | APPROVED FOR CONSTRUCTION | 황래중 | 안남후 | 박동영   | LINE NO.   | 40A-IW-ST1-C-9536                                                                     | SH'T NO.<br>01 / 02 | REV.<br>0 |
|            |        |    |           |                     |               |                                             |     | REV. NO.                                                                            | DATE     | DESCRIPTION               | BY  | CHK | APP'D |            |                                                                                       |                     |           |



